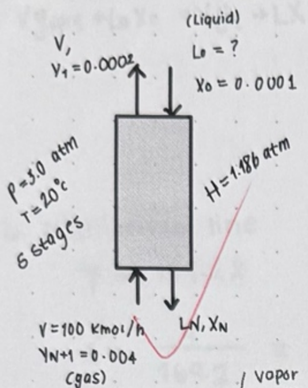
VIANCA ESCALANTE

12.D1



equilibrium curve

$$y = \frac{HX}{P_{tot}} = \frac{1.186}{3} x = \underbrace{0.395}_{M} x$$

comp mass balance

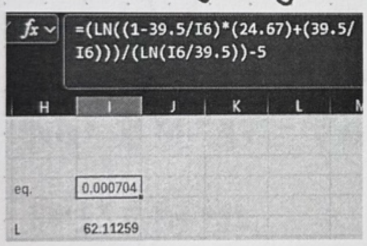
$$\underbrace{V y_{N+1}}_{\text{Known}} + \underset{\text{unk.}}{L_0} \underset{\text{Known}}{X_0} = V \underset{\text{Known}}{y_1} + L \underset{\text{unknown}}{X_N}$$

(vapor composition equili. w/ inlet liquid) $\longrightarrow y^* = (0.395)(0.0001) = 0.0000395$

Eq 12-22

$$5 = \frac{\ln\left[\left(1 - \frac{(0.395)(100)}{L}\right)\left(\frac{(0.004) - (0.0000395)}{(0.0002 - 0.0000395)}\right) + \frac{(0.395)(100)}{L}\right]}{\ln\left(\frac{L}{(0.395)(100)}\right)}$$

★ move everything to one side goal seek to equal zero



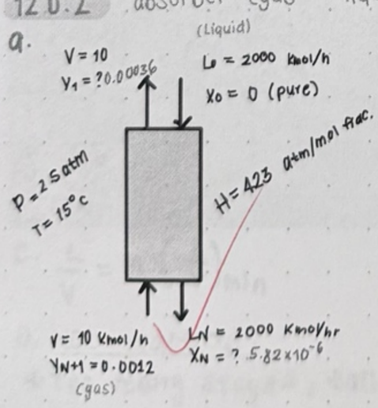
$\boxed{L = 62.11259 \text{ Kmol/hr}}$

$$X_N = \frac{LX_0 + Vy_{N+1} - Vy_1}{L} = \frac{(62.11259)(0.0001) + (100)(0.004) - (100)(0.0002)}{62.11259}$$

$\boxed{X_N = 0.0062}$

12.D.2 absorber (gas → liquid)

a.



97 % recovery H_2S in liquid meaning 3% left in gas

$$Y_{out} = (0.03)(0.0012) = \boxed{0.000036}$$

equilibrium line

$$y = \frac{H}{P} x = \frac{423}{2.5} x = 169.2 x$$

a:

$Vy_{N+1} + L_0 x_0 = Vy_1 + Lx_N \Rightarrow x_N = \dfrac{Vy_{N+1} + L_0 x_0 - Vy_1}{L}$

$$= \frac{(10)(0.0012) + (2000)(0) - (10)(0.00036)}{(2000)}$$

$$\boxed{= 5.82 \times 10^{-6}}$$

b. equilibrium line

$y = 169.2x$

$x^* = \dfrac{y}{169.2} = \dfrac{0.0012}{169.2} = 7.09 \times 10^{-6}$

op line (min)

value of x in equil. w/ y_{N+1}

has to go through $(x_0, y_1) = (0, 0.000036)$ & $(x_N^*, y_{N+1}) = (7.09\times10^{-6}, 0.0012)$

$\left(\dfrac{L}{V}\right)_{min} = \dfrac{y_{N+1} - y_1}{x_N^* - x_0} = \dfrac{0.0012 - 0.000036}{7.09\times10^{-6}} = 164.124$

$y = 164.124 + 0.000036$

op line

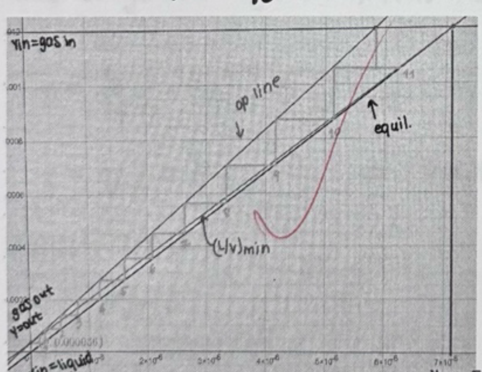
has to go through $(x_0, y_1) = (0, 0.000036)$ & $(x_0, y_1) = (5.82\times10^{-6}, 0.0012)$

$x = 0$

$y_{N+1} = \dfrac{L}{V} x_N + y_1 - \dfrac{L}{V} x_0$

$y_{N+1} = \dfrac{L}{V} x_N + y_1$

slope : $\dfrac{L}{V} = \dfrac{2000}{10} = 200 \Rightarrow y = 200x + 0.000036$



$\boxed{11 \text{ stages}, \sim 10.5 \text{ stages}}$

Xout = liquid equilibrium

c. $\dfrac{L}{V} = M\left(\dfrac{L}{V}\right)_{min}$ $\dfrac{L}{V} = \dfrac{2000}{10} = 200$ $200 = M(164.124) = \boxed{1.2186}$

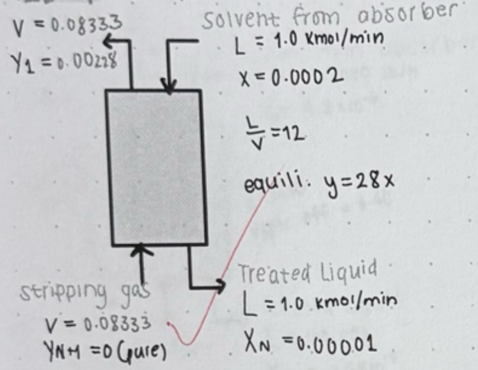
D. <u>Not practical:</u>

* Too many stages, tall column for small amount of solute (0.12%)

operating close to minimum solvent rate $(L/V)/(L/V)_{min} = 1.22$
H_2S is poorly absorbed big H value
* X_{out} very small (5.82×10^{-6}) for the amount of usage of water (2000 Kmol/hr)

Make practical:
* increase L/V, fewer stages needed
* use better solvent, increase in absorption efficiency
* Lower temp T↓ H↓
* increase pressure $y = H/p \cdot x$

12D.4 Stripper (liquid → gas)



vapor flow rate

$\frac{L}{V} = 12 \qquad V = \frac{1.0}{12} = 0.08333$ Kmol/min

balance

overall m.B: $V + L = V + L$

comp m.B: $y_{j+1}V + X_0 L = y_1 V + X_j L$

$y_{j+1} = 0$

$L X_0 = L X_j + V y_1 \Rightarrow y_1 = \frac{L X_0 - L X_j}{V} \Rightarrow y_1 = \frac{L}{V}(X_0 - X_j)$

$y_1 = 12(0.0002 - 0.00001)$

$\boxed{y_1 = 0.00228}$

has to go through $(X_{in}, Y_{in}) = (0.0002, 0.00228)$ & $(X_{out}, Y_{out}) = (0.00001, 0)$

op line

$y = \frac{L}{V}x + \left(y_{N+1} - \frac{L}{V}x_N\right) \qquad y_{in} = 0$

$y = \frac{L}{V}x - \frac{L}{V}x_N \Rightarrow y = 12x - 12(0.00001) \Rightarrow y = 12x - 0.00012$

equili. line

$y = 28x$

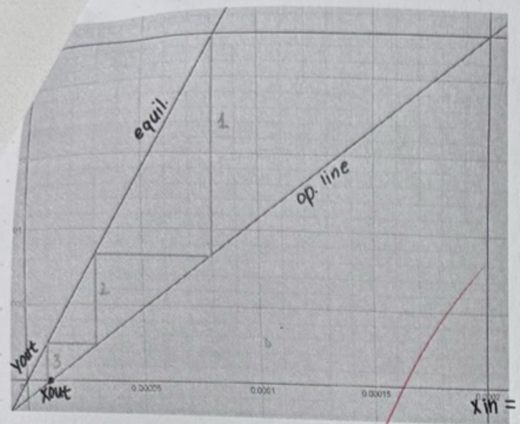
Check endpoints

@ $x = X_{out} = 0.00001$

$y = 12(0.00001) - 0.00012 = 0 \qquad (0.0001, 0)$ ✓

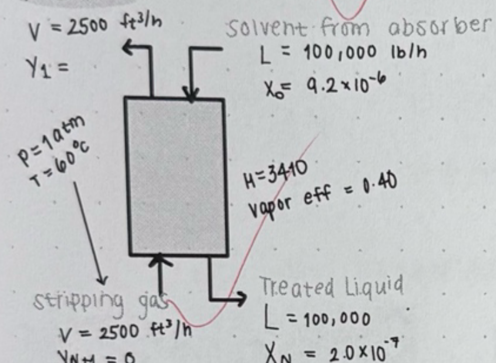
@ $x = X_{in} = 0.0002$

$y = 12(0.002) - 0.00012 = 0.00228 \qquad (0.0002, 0.00228)$ ✓



$\boxed{\approx 3 \text{ stages}}$

12D.6



$$y = \frac{H}{P_{tot}} x = \frac{3410}{1} x = \underbrace{3410}_{M} x$$

flow rates

MW of H_2O = 18.015

$$L = \frac{100{,}000}{18.015} = 5550.93 \text{ lbmol/hr}$$

$$V = \frac{PV}{RT} = \frac{(1)(2500 \text{ ft}^3/\text{h})}{(0.7302 \frac{\text{atm ft}^3}{\text{lbmol} \cdot R})(599.67 R)} = 5.709 \text{ lbmol/h} \longrightarrow \frac{L}{V} = \frac{5550.93}{5.709} = 972.3$$

balance

overall m.B: $V + L = V + L$

comp m.B: $y_{N+1} V + x_0 L = y_1 V + x_N L$

$$y_1 = \frac{x_0 L - x_N L}{V} = \frac{L}{V}(x_0 - x_N) = 972.3(9.2\times10^{-6} - 2.0\times10^{-7}) = \boxed{0.00875}$$

Eq 12-34

$m\frac{V}{L} = (3410)(0.00103) = 3.508$ $\quad y_{N+1} = 0 \quad y_1 = 0.00875$

$y_1^* = (3410)(9.2\times10^{-6}) = 0.03137$ $\quad E_{MV} = 0.40$

$$N_{actual} = -\frac{\ln\{[1-3.508][(0-0.03137)/(0.00875-0.03137)]+3.508\}}{\ln[1+(0.40)(3.508-1)]}$$

$$= \boxed{5.07 \text{ Stages}}$$

* use 6 stages because effiency has to be accurate